NATIONAL OPEN UNIVERSITY OF NIGERIA
PLOT 91, CADASTRAL ZONE, NNAMDI AZIKIWE EXPRESSWASY, JABI-ABUJA
FACULTY OF SCIENCES
DEPARTMENT OF COMPUTER SCIENCE
JANUARY, 2021 EXAMINATIONS

Course Code: CIT383
Course Title: INTRODUCTION TO OBJECT-ORIENTED PROGRAMMING
Credit: 2
Time Allowed: 2 hours
INSTRUCTION: Answer question one and any other three (3) questions

QUESTION ONE (25MARKS) COMPULSORY

a. Describe Polymorphism and identify it major types (5marks)
b. Identify the examples of message passing styles (5marks)
c. Write the syntax for importing individual classes and all classes in a package respectively (5marks)
d. Describe the term Data hiding (5marks)
e. Identify two categories of Data type (2marks)
f. Differentiate between constructor and method (3marks)

QUESTION TWO

a. Write a Java class to print the message "I love Java Programming" (6marks)
b. Describe the function of "association property" of a class and identify two advantages of software reusability (6marks)
c. Outline two functions of a destructor (3marks)

QUESTION THREE

a. Outline and describe the motivations for modularising a program into methods (6marks)
b. Show the comparison of conversion between implicit and explicit casts of numeric data (100) using sample code (6marks)
c. Identify the components of a method signature (3marks)

QUESTION FOUR

a. Demonstrate how a main method is declared and show the syntax of creating method (6marks)
b. Describe the function of a method and summarize any 4 advantages of using method (7marks)
c. What is ADT? (2marks)

QUESTION FIVE

a. Describe the term Recursive Methods (5marks)
b. State two limitations of overloading True and False operators (2marks)
c. Differentiate between an implicit cast and an explicit cast (3marks)
d. Explain the terms array and indexer (5marks)